COMPUTER ENGINEERING

QUEZON CITY

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| **COURSE CODE/TITLE:** | CPE 212 - Automating Server Management | **SECTION:** | CPE31s2 |
| **DATE PERFORMED:** | 13/08/2026 | **DATE SUBMITTED:** | 13/08/2026 |
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**ACTIVITY NO. ___ Prelim Exam**

## 1. PROCEDURE OUTPUT

What procedures did you perform?

Replace the sample IP addresses below with the actual IP addresses of your virtual machines.

### 1. Set up

| Hostname | Operating System | Ip Address |
|---|---|---|
| Jakews | Ubuntu | 192.168.85.4 |
| jakeserver1 | Ubuntu | 192.168.85.5 |
| jakeserver2 | Ubuntu | 192.168.85.6 |
| JakeCentos | CentOs | 192.168.85.8 |

### Part A: Installing and Verifying Ansible

```
jakews@workstation:~$ ansible --version
ansible 2.10.8
  config file = None
  configured module search path = ['/home/jakews/.ansible/plugins/modules', '/us
r/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  executable location = /usr/bin/ansible
  python version = 3.10.12 (main, Jun 22 2026, 18:55:27) [GCC 11.4.0]
jakews@workstation:~$
```

```
jakews@workstation:~$ ssh jakews@192.168.85.6
jakews@192.168.85.6's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-136-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

13 updates can be applied immediately.
13 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu Aug 13 08:15:04 2026 from 192.168.85.4
jakews@server2:~$
```

```
jakews@workstation:~$ ssh jakews@192.168.85.5
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-136-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Last login: Fri Jul 31 12:44:59 2026 from 192.168.85.4
jakews@server1:~$
```

```
Connection to 192.168.85.6 closed.
jakews@workstation:~$ ssh jgestiada@192.168.85.8
The authenticity of host '192.168.85.8 (192.168.85.8)' can't be established.
ED25519 key fingerprint is SHA256:ucuenmeGi4W70VCNpqjVDk/LCGjNhiCHnZaDl+voqDQ.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.85.8' (ED25519) to the list of known hosts.
jgestiada@192.168.85.8's password:
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Thu Aug 13 07:55:01 2026
[jgestiada@jakecentos ~]$
```

**Part B: Building an Inventory File**

```
jakews@workstation: ~/ansible-lab
  GNU nano 6.2                          inventory.yaml
all:
  children:

    ubuntu:
      hosts:
        ubuntu1:
          ansible_host: 192.168.85.5

        ubuntu2:
          ansible_host: 192.168.85.6


    centos:
      hosts:
        centos1:
          ansible_host: 192.168.85.8
```

```
jakews@workstation:~/ansible-lab$ ansible-inventory -i inventory.yaml --list
{
    "_meta": {
        "hostvars": {
            "centos1": {
                "ansible_host": "192.168.85.8"
            },
            "ubuntu1": {
                "ansible_host": "192.168.85.5"
            },
            "ubuntu2": {
                "ansible_host": "192.168.85.6"
            }
        }
    },
    "all": {
        "children": [
            "centos",
            "ubuntu",
            "ungrouped"
        ]
    },
    "centos": {
        "hosts": [
            "centos1"
        ]
    },
    "ubuntu": {
        "hosts": [
            "ubuntu1",
            "ubuntu2"
        ]
    }
```

```
server2 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
server1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
[DEPRECATION WARNING]: Distribution centos 9 on host server3 should use
/usr/libexec/platform-python, but is using /usr/bin/python for backward
compatibility with prior Ansible releases. A future Ansible release will
default to using the discovered platform python for this host. See https://docs
.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html for
more information. This feature will be removed in version 2.12. Deprecation
warnings can be disabled by setting deprecation_warnings=False in ansible.cfg.
server3 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python"
    },
    "changed": false,
    "ping": "pong"
```

## Part C: Creating a Metagroup

```
jakews@workstation:~/ansible-lab$ ansible all -i inventory.yaml --list-hosts
  hosts (3):
    server1
    server2
    server3
```

Trash

```
jakews@workstation:~/ansible-lab$ ansible all -i inventory.yaml -m ping
server2 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
server1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
[DEPRECATION WARNING]: Distribution centos 9 on host server3 should use /usr/libexec/platform-python
 using /usr/bin/python for backward compatibility with prior Ansible releases. A future Ansible rele
 default to using the discovered platform python for this host. See
https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html for more infor
This feature will be removed in version 2.12. Deprecation warnings can be disabled by setting
deprecation_warnings=False in ansible.cfg.
server3 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python"
    },
    "changed": false,
    "ping": "pong"
}
```

## Part D: Creating Variables

```
jakews@workstation:~/ansible-lab$ ansible-playbook -i inventory.yaml testvars.yaml

PLAY [Display Variables] ***************************************************************************

TASK [Gathering Facts] *****************************************************************************
ok: [server1]
ok: [server2]
[DEPRECATION WARNING]: Distribution centos 9 on host server3 should use /usr/libexec/platform-python, but is
 using /usr/bin/python for backward compatibility with prior Ansible releases. A future Ansible release will
 default to using the discovered platform python for this host. See
https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html for more information.
This feature will be removed in version 2.12. Deprecation warnings can be disabled by setting
deprecation_warnings=False in ansible.cfg.
ok: [server3]

TASK [Show Operating System] ***********************************************************************
ok: [server1] => {
    "msg": "Host server1 is running Ubuntu"
}
ok: [server2] => {
    "msg": "Host server2 is running Ubuntu"
}
ok: [server3] => {
    "msg": "Host server3 is running CentOs"
}

PLAY RECAP *****************************************************************************************
server1                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server2                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server3                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

## Part E: Creating Your First Playbook


```
jakews@workstation:~/ansible-lab$ ansible-playbook -i inventory.yaml testvars.yaml

PLAY [Display Variables] ***************************************************************************

TASK [Gathering Facts] *****************************************************************************
ok: [server1]
ok: [server2]
[DEPRECATION WARNING]: Distribution centos 9 on host server3 should use /usr/libexec/platform-python, but is
 using /usr/bin/python for backward compatibility with prior Ansible releases. A future Ansible release will
 default to using the discovered platform python for this host. See
https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html for more information.
This feature will be removed in version 2.12. Deprecation warnings can be disabled by setting
deprecation_warnings=False in ansible.cfg.
ok: [server3]

TASK [Show Operating System] ***********************************************************************
ok: [server1] => {
    "msg": "Host server1 is running Ubuntu"
}
ok: [server2] => {
    "msg": "Host server2 is running Ubuntu"
}
ok: [server3] => {
    "msg": "Host server3 is running CentOs"
}

PLAY RECAP *****************************************************************************************
server1                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server2                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server3                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

```
jakews@workstation:~/ansible-lab$ ansible-playbook -i inventory.yaml playbook.yaml

PLAY [First Ansible Playbook] *************************************************

TASK [Gathering Facts] ********************************************************
ok: [server1]
ok: [server2]
[DEPRECATION WARNING]: Distribution centos 9 on host server3 should use
/usr/libexec/platform-python, but is using /usr/bin/python for backward compatibility
with prior Ansible releases. A future Ansible release will default to using the
discovered platform python for this host. See
https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html
for more information. This feature will be removed in version 2.12. Deprecation
warnings can be disabled by setting deprecation_warnings=False in ansible.cfg.
ok: [server3]

TASK [Ping all managed nodes] *************************************************
ok: [server1]
ok: [server2]
ok: [server3]

TASK [Display Hello World] ****************************************************
ok: [server1] => {
    "msg": "Hello World from server1"
}
ok: [server2] => {
    "msg": "Hello World from server2"
}
ok: [server3] => {
    "msg": "Hello World from server3"
}

TASK [Display Operating System] ***********************************************
ok: [server1] => {
    "msg": "Operating system: Ubuntu "
}
ok: [server2] => {
    "msg": "Operating system: Ubuntu "
}
ok: [server3] => {
    "msg": "Operating system: CentOs "
```

## 2. ANSWERS TO SUPPLEMENTAL ACTIVITY

```
jakews@workstation:~/ansible-lab$ ansible-playbook -i inventory.yaml challenge.yaml

PLAY [Supplementary] ***********************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [server1]
[DEPRECATION WARNING]: Distribution centos 9 on host server3 should use /usr/libexec/platform-python, but is using
/usr/bin/python for backward compatibility with prior Ansible releases. A future Ansible release will default to using the
discovered platform python for this host. See
https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html for more information. This feature
will be removed in version 2.12. Deprecation warnings can be disabled by setting deprecation_warnings=False in ansible.cfg.
ok: [server3]

TASK [Display Message] *********************************************************************************************
ok: [server1] => {
    "msg": "Welcome to Ansible Laboratory"
}
ok: [server3] => {
    "msg": "Welcome to Ansible Laboratory"
}

PLAY RECAP *********************************************************************************************************
server1                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server3                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

jakews@workstation:~/ansible-lab$
```

```
Firefox Web Browser        jakews@workstation: ~/ansible-lab
  GNU nano 6.2                                challenge.yaml
---
- name: Supplementary
  hosts: servers

  tasks:
    - name: Display Message
      debug:
        msg: "Welcome to Ansible Laboratory"
```

Copy and paste the questions from the lab and include your answers after.

Guide Questions

1. What is the role of the Control Node in Ansible?
   - This is the central use of hosts that can manage the automation task, playbook in the server that will activate to execute them remotely.
2. Why is an inventory file required?
   - The inventory file tells Ansible which servers to manage and allows us to group them.
3. What is the purpose of a metagroup?
   - A metagroup combines different groups of servers so we can manage them together. This command for the vars changes.
4. How do variables simplify Ansible administration?
   - Variables store values like IP addresses, usernames, and settings, making playbooks to reuse and manage.
5. What is the difference between an ad hoc command and a playbook
   - An ad hoc command is used for a quick, one-time task, and a playbook contains organized tasks that can be run repeatedly.

Conclusion:

- In this activity, we setup the servers in the first part then use a workstation for ansible that can connect to the 3 server ssh in ubuntu and centOS. The Part B is to manage to set up on yaml to connect servers IP address ssh this would verify the connection of servers ssh. The /etc/hosts need to command the servers and ip address to check the -list-host shown in verify. Then I learned how Ansible can make server

management by allowing one control node to manage multiple machines. I also learned how inventory files, groups, variables, and playbooks work together to automate tasks.

## 3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

**Tools Used:** _____

*Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.*

**Purpose of Use:** _____

*Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.*

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